Practical thermometry

- Zero-field
  - Platinum 77-300 K
  - Cernox (zirconium-axynitride) 0.5-77 K
  - Ruthenium oxide (20 mK - 0.5 K)
  - (simplicity – silicon diode (1.5 – 300 K)
- In a magnetic field
  - Cernox (various models – check the specific characteristics) 0.3-325 K
  - Platinum 77-300 K
  - Capacitance 1- 290 K

Installation

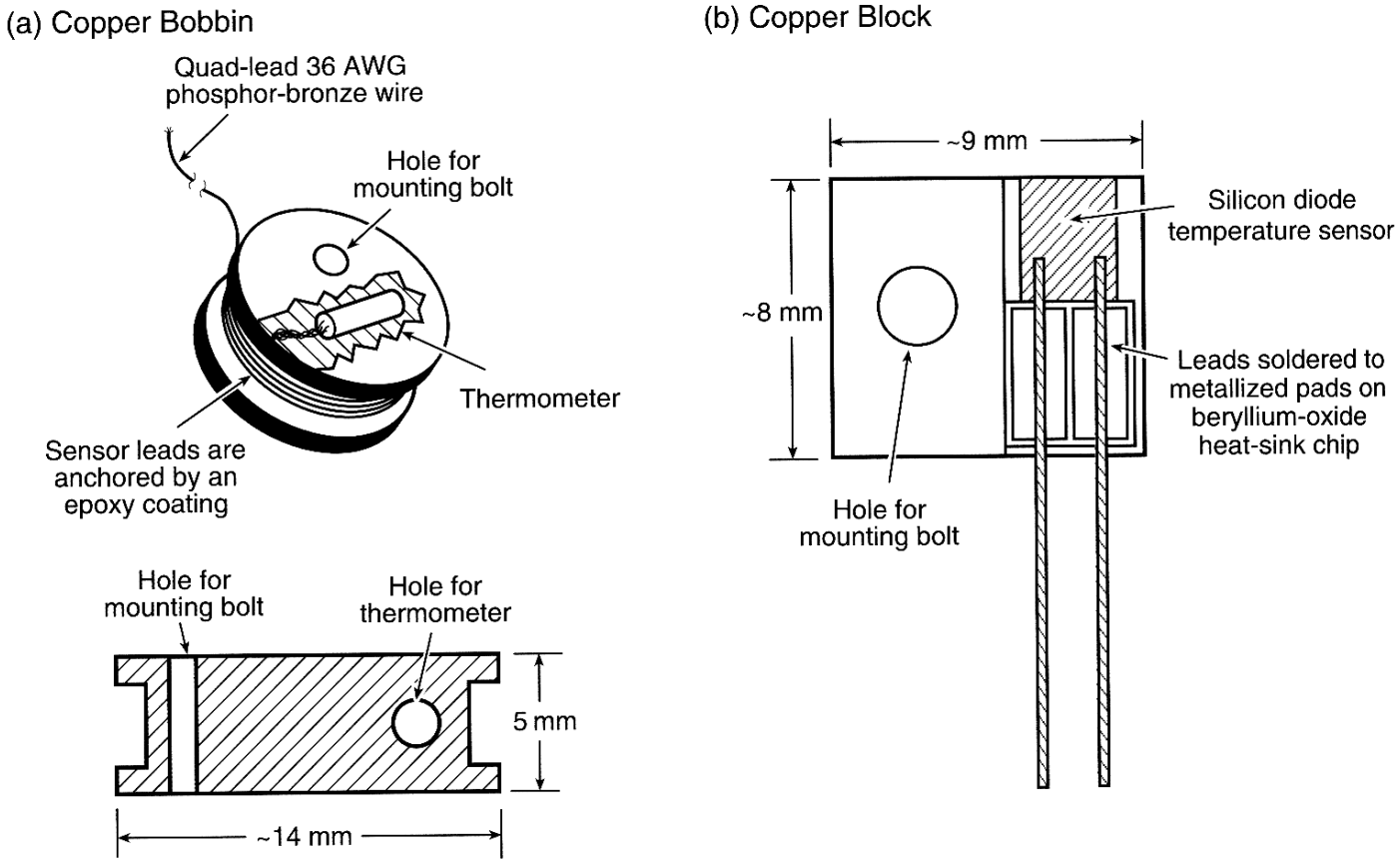


Fig. 5.12 Bolt-on copper bobbin (a), and copper block (b), for thermally anchoring a thermometer and its leads (adapted from Lake Shore Cryotronics 2002). For the bobbin technique, the thermometer leads are wrapped around the bobbin and thermally attached to it with varnish or epoxy. For the copper block technique, a beryllium-oxide chip with electric-terminal metallization on top can be used to thermally anchor the thermometer leads. To enhance the thermal contact between the bobbin/block and sample holder, a thin layer of thermally conductive grease is applied to the bottom of the bobbin/block before it is bolted in place.

Thermal, but not electrical contact

Varnish, such as IMI 7031 varnish (formerly GE 7031), can be painted directly onto the copper sample holder to provide an electrically insulating layer. Application is easier if the varnish is first partially thinned with ethanol. Normal varnish curing time (12–24 h) can be shortened by using a heat lamp. If the varnish needs to be removed later, the same thinner can be used. This method is well suited to large or long samples, since large areas can be readily painted.

Cigarette paper can add some robustness to varnish (the paper is available commercially). This method is appropriate for situations where the soft varnish layer might be scratched. Thoroughly soak the paper with thinned varnish to adhere the thin paper to the copper sample holder as well as to provide good thermal conduction through the paper.

Mylar™- or Kapton™-based tape is perhaps the easiest to install. Simply stick it to the copper sample holder. "Yellow" Mylar™ tape (3M #56) is very thin, and its adhesion to copper actually improves with thermal cycling (application notes and supplier information are given in the table of sticky stuff, Appendix A3.10). Thin, adhesive-coated Kapton™ tape also works well. Both types of tape are more robust than a varnish layer alone.

Thin pieces of *sapphire* or *silicon wafer* can be greased to the sample holder. This method works best for electrically isolating small samples that do not require large sample holders.

Thermometer location

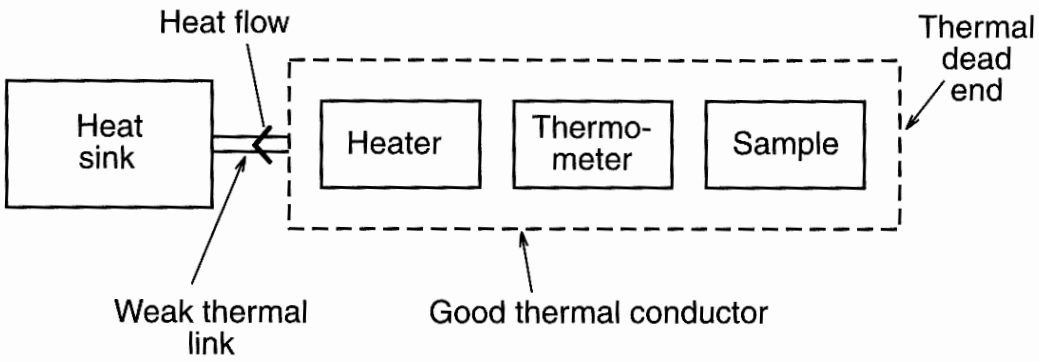
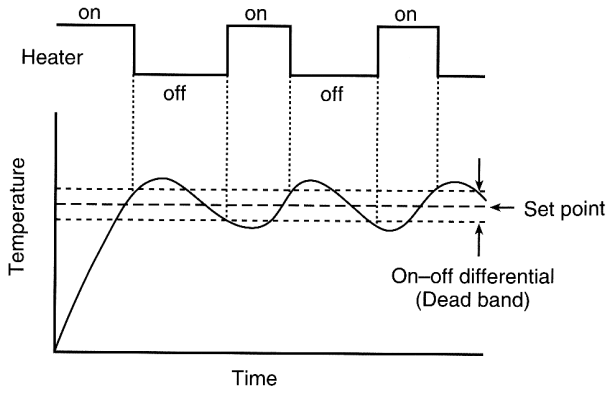


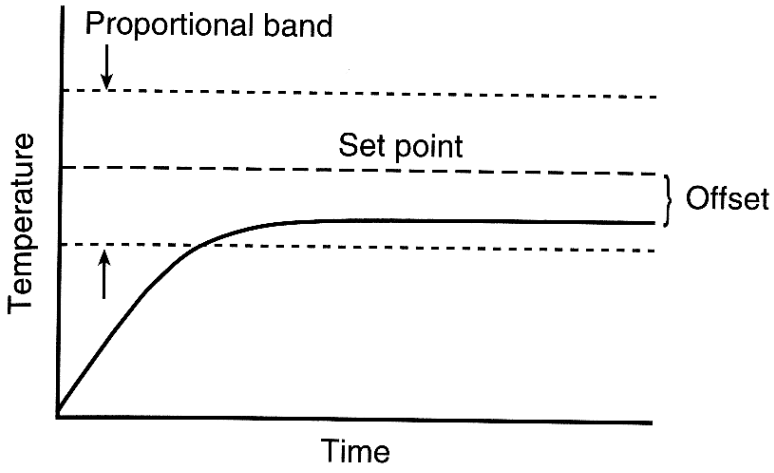
Fig. 5.13 Ideal arrangement for thermometer location.

Temperature control

- on-off controllers



- proportional



Proportional tuning involves correcting a target proportional to the difference. Thus, the target value is never achieved because as the difference approaches zero, so too does the applied correction.

Integral tuning attempts to remedy this by effectively cumulating the error result from the "P" action to increase the correction factor. For example, if the oven remained below temperature, “I” would act to increase the head delivered. However, rather than stop heating when the target is reached, "I" attempts to drive the cumulative error to zero, resulting in an overshoot.

Derivative tuning attempts to minimize this overshoot by slowing the correction factor applied as the target is approached.

- proportional-integral-derivative

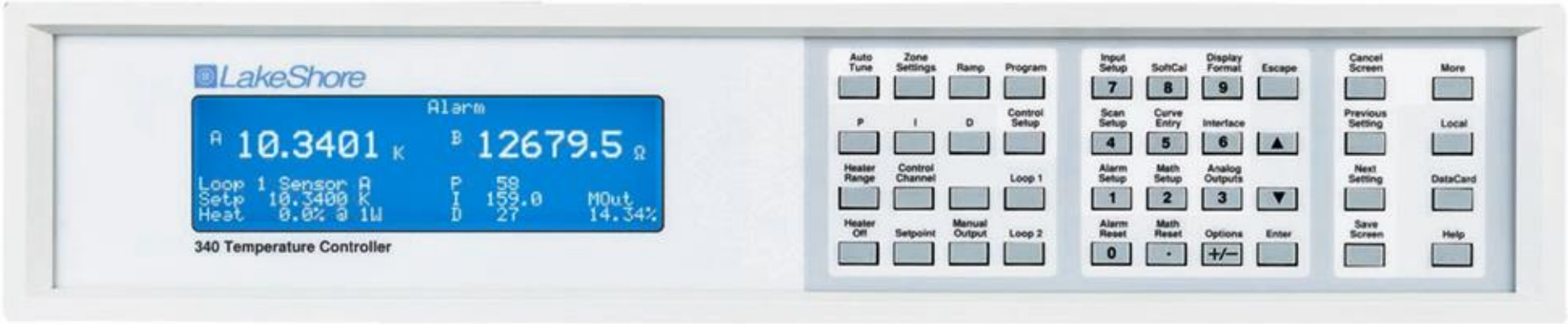
A proportional integral derivative (PID) controller can be used as a means of controlling temperature, pressure, flow and other process variables. As its name implies, a PID controller combines proportional control with additional integral and derivative adjustments which help the unit automatically compensate for changes in the system.

When selecting a commercial temperature controller, it pays to check the following:

- *Resolution and absolute accuracy*: Is it adequate to meet your temperature-measurement needs?
- *Input modules*: Are they available for the particular sensors you want to use?
- *Sensor excitation level*: Can it be set low enough to avoid self-heating for a particular sensor and temperature range?
- *ac sensor excitation*: Is it available to avoid the effects of slowly drifting thermoelectric voltages?
- *Interpolation schemes*: What types are used to fit the calibration data? (See suggestion list in Sec. 5.1.6.) Can new calibrations be entered?
- *Preloaded calibrations*: Are they available for nonprecision work with interchangeable thermometers?
- *Heater power*: Is it adequate and a genuine analogue output (not pulsed, which causes electromagnetic interference for precise work)?
- *Computer interface*: Is it available? Is it electrically isolated from the cryogenic outputs to minimize interference?


LakeShore
370 AC Resistance Bridge
49.506 kΩ
48.012 mK
Setpoint 48.000 mK
Heat 20µW of 100 µW
Range 100 kΩ
Excitation 10 µV
Sensor Power 495 aW
Autorange On/Off
Voltage Mode
Current Mode
Autoscan On/Off
Resistance Range
Voltage Excitation
Current Excitation
Scanner Channel
Zone Setup
Control Setup
Manual Output
Still Output
PID
Heater Range
Setpoint
Heater Off
Input Setup
Display Setup
Local
Escape
7
8
9
Instrument Setup
Curve Entry
Computer Interface
4
5
6
Filter
Linear (ΔR)
Analog Output
1
2
3
Max/Min
Alarm
Relay
Enter
0
.
+/−


RS-232 (DTE)
IEEE-488 INTERFACE
ALARM RELAYS
ANALOG OUT 2 (STILL)
ANALOG OUT 1
HEATER OUTPUT
MONITOR
REFERENCE
SCAN CONTROL
INPUT
V+
V-
VG+
VG-
SHIELD
I+
I-
IG+
IG-
SHIELD





❶ - *Line Input Assembly*
❷ - *Heater Fuse*
❸ - *Heater Output*
❹ - *Option Slots*
❺ - *Data Card*
❻ - *IEEE-488 Interface*
❼ - *Serial (RS-232C) I/O*
❽ - *Digital I/O*
❾ - *Relays*
❿ - *Analog Outputs*
⓫ - *Standard Sensor Inputs*


LakeShore
5.2014
27.8645
4.6103
35.6387
336 Temperature Controller
A
B
C
D
SETPOINT
Heater Range
P
I
D
Manual Out
CONTROL OUTPUTS
ALL OFF
7
8
9
ESCAPE
4
5
6
1
2
3
0
.
+/-
ENTER
Remote
Ethernet
Alarm

Lake Shore offers a complete line of cryogenic accessories for sensor installation and general-purpose cryogenic use.

Cryogenic Wire and Cable

Used to minimize heat leak into the sensor and cryogenic system, cryogenic wire has a much lower thermal conductivity (and higher electrical resistivity) than copper wire.

The most common type of cryogenic wire is phosphor bronze. This wire is available in one-, two-, and four-lead configurations. Four-lead configurations are available as Quad-twist™ (two twisted pairs) or Quad-lead™ (ribbon). Wire gauge is 32 or 36 AWG, with polyimide or polyvinyl formal (Formvar®) used to insulate the wires.

Other common cryogenic wires and coaxial cables include manganin, nichrome heater wire, and HD-30 heavy duty copper wire. For high-frequency signals, Lake Shore provides various coaxial cables: ultra miniature coaxial cables and semi-rigid coaxial with a stainless steel center conductor.

Solder

The most common electrical connections are solder joints. Solder can also be used to install various sensors to improve thermal heat sinking. Common solders are indium solder and 90/10 Pb/Sn. Indium solder is used for various applications including sensor installation to provide excellent thermal contact with the sample. 90/10 Pb/Sn solder is used for applications requiring a higher temperature (liquidus point of 575 K and solidus point 458 K). Ostalloy® 158 solder is used as a seal for demountable vacuum cans and electric feedthroughs in cryogenic systems.

Epoxy, Thermal Grease, and Varnish

Thermal greases and epoxies are used to install and fasten sensors, while providing thermal contact and/or electrical insulation, with the sample. Epoxy can be used for mechanical attachment and joints.

The most common varnish for cryogenic installations is VGE-7031 varnish. It has good chemical resistance, bonds to a variety of materials, and has a fast tack time. Stycast® 2850FT is composed of a black epoxy resin, and has a thermal expansion coefficient that is matched to copper. A silver-filled, low-temperature conducting epoxy provides excellent strength, along with electrical and thermal conductivity.

Thermal grease, Apiezon N and Apiezon H, is suitable for enhancing thermal contact, especially for sensors inserted into cavities. Apiezon N is for low temperature applications, while H is for high temperature.

Miscellaneous Cryogenic Accessories

Lake Shore also supplies heat sink bobbins, a beryllium oxide heat sink chip, and a four-lead resistance sample holder. Cartridge heaters and vacuum feed through products are also available.

Table 5.1 **Properties of selected cryogenic thermometers (~1–~300 K).**
A more complete listing of cryogenic thermometers and their properties is given in the addendum at the end of this chapter (Sec. 5.5) and in Appendix A5.2.

Sensor Type	Temperature Range	Accuracy[a] (± value)	Reproducibility[b] (± value)	Long-Term Calibration Drift	Inter-change-ability[c]	Magnetic Field Use	Best Use	Cost
Platinum resistance thermometer	77–800 K With impurity correction: 20–77 K (Appendix A5.3b)	Without individual calibration: 0.6 K at 70 K 0.2 K at 300 K With individual calibration: 20 mK at 70 K 35 mK at 300 K	10 mK from 77 K to 305 K	±10 mK/yr at 77 K to 237 K	Yes	Recommended above 70 K; error < 0.1% with standard correction given in Appendix A5.5	Measurements above 77 K Excellent reproducibility interchangeability, low magnetic field error Many shapes and sizes available	Low without calibration High with individual calibration
Zirconium–oxynitride resistance thermometer (Cernox™)	0.3–325 K	Must be individually calibrated 5 mK at 4.2 K < 0.1% at > 10 K	3 mK at 4.2 K	±25 mK/yr at 1 K to 100 K 0.05% of reading at 100 K to 300 K	No	Recommended Lowest error Standard correction given Appendix A5.6	One of the best sensors for use in magnetic fields Good sensitivity over a wide temperature range Fast response time as chip	High with individual calibration
Germanium resistance thermometer	0.05–100 K	Must be individually calibrated With individual calibration: 5 mK at < 10 K 0.07% at > 10 K	0.5 mK at 4.2 K	±1 mK/yr at 4.2 K ±10 mK/yr at 77 K	No	Not recommended	Secondary-standard thermometer Excellent reproducibility	High with individual calibration
Silicon diode thermometer	1.4–450 K	Without calibration: 1 K at < 100 K 1% at 100 K to 300 K With individual calibration: 20 mK at 1.4 K–10 K 50 mK at 10 K–330 K	5 mK at 4.2 K 20 mK at 77 K 15 mK at 300 K	±10 mK/yr at 4.2 K ±40 mK/yr at 77 K ±25 mK/yr at 300 K	Yes	Not recommended below ~60 K	Relatively inexpensive, interchangeable, easily measured output Small size	Medium for low accuracy High with individual calibration

* Compiled from data supplied by Lake Shore Cryotronics 2002, Besley (1993), Rubin (1999).
[a] Accuracy: the difference between the measured and true temperature value.
[b] Reproducibility: the change in apparent temperature when the sensor is subjected to repeated thermal cycling from room temperature.
[c] Interchangeability: the ability to substitute one sensor for another with little change in calibration.